



WEB DESIGN & DEVELOPMENT

ASSIGNMENT



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Course: Web Design & Development

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ASSIGNMENT#2

CALCULATOR

HTML code:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Simple Calculator</title>
5   <link rel="stylesheet" href="calculator.css">
6 </head>
7 <body>
8   <div class="calculator">
9     <input type="text" id="display" disabled>
10    <br>
11    <button class="btn" onclick="appendNumber('7')">7</button>
12    <button class="btn" onclick="appendNumber('8')">8</button>
13    <button class="btn" onclick="appendNumber('9')">9</button>
14    <button class="btn" onclick="appendOperator('/')">/</button>
15    <br>
16    <button class="btn" onclick="appendNumber('4')">4</button>
17    <button class="btn" onclick="appendNumber('5')">5</button>
18    <button class="btn" onclick="appendNumber('6')">6</button>
19    <button class="btn" onclick="appendOperator('*')">*</button>
20    <br>
21    <button class="btn" onclick="appendNumber('1')">1</button>
22    <button class="btn" onclick="appendNumber('2')">2</button>
23    <button class="btn" onclick="appendNumber('3')">3</button>
24    <button class="btn" onclick="appendOperator('-')">-</button>
25    <br>
26    <button class="btn" onclick="appendNumber('0')">0</button>
27    <button class="btn" onclick="appendNumber('.')">.</button>
28    <button class="btn" onclick="calculateResult()">=</button>
29    <button class="btn" onclick="appendOperator('+')">+</button>
30    <br>
31    <button class="btn" onclick="clearDisplay()">C</button>
32  </div>
```

```
34 <script src="calculator.js"></script>
35 </body>
36 </html>
```

Code Details – Calculator (HTML)

Document Declaration

- The **DOCTYPE** declaration defines the document as **HTML5**.
-

HTML Root Element

- The **html** tag contains the complete structure of the webpage.
-

Head Section

- The **head** section stores information about the webpage.
 - The **title** tag sets the webpage title as *Simple Calculator*.
 - The **link** tag connects the external **calculator.css** file for styling.
-

Body Section

- The **body** tag contains all visible content of the calculator.

Calculator Container

- A **div** with class **calculator** is used as the main container.
 - It helps group all calculator elements together for styling.
-

Display Input Field

- The **input** field displays numbers and results.
 - **type="text"** allows text display.
 - **id="display"** is used to access it in JavaScript.
 - **disabled** prevents the user from typing manually.
-

Number Buttons

- **button** elements are used for numbers (0–9).
 - **onclick** calls the **appendNumber()** function.
 - The clicked number is added to the display.
-

Operator Buttons

- Operator buttons **+**, **-**, **×**, **/** are created using buttons.
 - **onclick** calls the **appendOperator()** function.
 - The selected operator is added to the display.
-

Equal Button

- The **=** button calls the **calculateResult()** function.
 - This function calculates and displays the result.
-

Clear Button

- The **C** button calls the **clearDisplay()** function.

- It clears the calculator display.

JavaScript File Linking

- The **script** tag links the external **calculator.js** file.
- JavaScript controls all calculator operations.

Conclusion

- This HTML code creates the **structure of a simple calculator**.
- It uses **buttons**, **input field**, and **JavaScript functions**.
- The calculator performs basic arithmetic operations in an easy way.

CSS code:

```
1  /* Body & background */
2  body {
3      font-family: Arial, sans-serif;
4      background-color: #1e1e2f;
5      display: flex;
6      justify-content: center;
7      align-items: center;
8      height: 100vh;
9  }
10
11 /* Calculator container */
12 .calculator {
13     background-color: #2b2b3d;
14     padding: 25px;
15     border-radius: 12px;
16     box-shadow: 0 6px 20px rgba(0,0,0,0.3);
17 }
18
19 /* Display at top */
20 #display {
21     width: 100%;
22     height: 50px;
23     font-size: 22px;
24     margin-bottom: 15px;
25     text-align: right;
26     padding-right: 10px;
27     border-radius: 6px;
28     border: 1px solid #444;
29     background-color: #1e1e2f;
30     color: #ffffff;
31 }
```

```
33 /* Buttons */
34 .btn {
35     width: 60px;
36     height: 50px;
37     margin: 5px;
38     font-size: 20px;
39     cursor: pointer;
40     border: none;
41     border-radius: 6px;
42     color: white;
43     background-color: #5c8cff;
44     transition: 0.2s;
45 }
46
47 /* Button hover effect */
48 .btn:hover {
49     background-color: #4a7be0;
50 }
51
52 /* Special clear button */
53 .btn-clear {
54     background-color: #ff5c5c;
55 }
56
57 .btn-clear:hover {
58     background-color: #e04a4a;
59 }
60
61 /* Container alignment for rows */
62 .row {
63     display: flex;
64     justify-content: center;
```

CSS Code Details – Calculator

Body and Background Styling

- The **body** selector applies styles to the whole page.
- **font-family** sets the text style to **Arial**.

- **background-color** gives the page a **dark theme**.
 - **display: flex** enables **flexbox layout**.
 - **justify-content: center** aligns the calculator **horizontally**.
 - **align-items: center** aligns the calculator **vertically**.
 - **height: 100vh** makes the layout cover the full screen height.
-

Calculator Container Styling

- The **.calculator** class styles the calculator box.
 - **background-color** gives a slightly lighter dark shade.
 - **padding** adds inner spacing around elements.
 - **border-radius** rounds the corners for a smooth look.
 - **box-shadow** adds a shadow for a **3D effect**.
-

Display Styling

- The **#display** selector styles the calculator screen.
 - **width: 100%** makes it span the full container width.
 - **height** increases display visibility.
 - **font-size** enlarges the displayed numbers.
 - **text-align: right** aligns numbers like a real calculator.
 - **padding-right** adds spacing from the edge.
 - **background-color** and **color** improve contrast.
-

Button Styling

- The **.btn** class styles all calculator buttons.
- **width** and **height** define button size.
- **margin** creates space between buttons.
- **font-size** increases button text size.

- **cursor: pointer** shows the hand cursor on hover.
 - **border-radius** smoothens button edges.
 - **background-color** applies a soft blue color.
 - **transition** makes hover effect smooth.
-

Button Hover Effect

- The **.btn:hover** selector changes button color on hover.
 - This gives **visual feedback** to the user.
-

Clear Button Styling

- The **.btn-clear** class styles the clear button separately.
 - **background-color** gives a red color to indicate reset action.
 - **hover effect** darkens the color when hovered.
-

Button Row Alignment

- The **.row** class uses **display: flex** to arrange buttons in a row.
 - **justify-content: center** aligns buttons neatly.
-

Java script code:

```
1 // Get display input
2 let display = document.getElementById("display");
3 // Variables to store numbers and operator
4 let num1 = "";
5 let operator = "";
6 let num2 = "";
7 // Function to append number
8 function appendNumber(number) {
9   display.value += number; // show on display
10  if (operator === "") {
11    num1 += number; // build first number
12  } else {
13    num2 += number; // build second number
14  }
15 }
16 // Function to append operator
17 function appendOperator(op) {
18   if (num1 === "") return; // don't allow operator first
19   if (operator !== "") return; // don't allow multiple operators
20   operator = op;
21   display.value += op;
22 }
23 // Function to calculate result
24 function calculateResult() {
25   if (num1 === "" || operator === "" || num2 === "") return; // prevent error
26   let result = 0;
27   let n1 = parseFloat(num1);
28   let n2 = parseFloat(num2);
29   if (operator === "+") result = n1 + n2;
30   else if (operator === "-") result = n1 - n2;
31   else if (operator === "*") result = n1 * n2;
32   else if (operator === "/") result = n1 / n2;
```

```
33   display.value = result;
34   // reset for next calculation
35   num1 = result.toString();
36   operator = "";
37   num2 = "";
38 }
39 // Function to clear display
40 function clearDisplay() {
41   display.value = "";
42   num1 = "";
43   operator = "";
44   num2 = "";
45 }
```

JavaScript Code Details – Simple Calculator

Selecting the Display

- **document** refers to the HTML page.
 - **getElementById("display")** selects the calculator display input field.
 - The **display** variable is used to show numbers and results.
-

Declaring Variables

- **num1** stores the **first number** entered by the user.
 - **operator** stores the **mathematical operator** (+, -, ×, ÷).
 - **num2** stores the **second number**.
 - These variables help control calculator logic.
-

appendNumber() Function

- This function runs when a **number button** is clicked.
 - **display.value += number** shows the clicked number on the screen.
 - **if (operator === "")** checks if an operator is not selected yet.
 - **num1 += number** builds the first number digit by digit.
 - **else** builds the second number using **num2 += number**.
-

appendOperator() Function

- This function runs when an **operator button** is clicked.
- **if (num1 === "") return** prevents operator before a number.
- **if (operator !== "") return** prevents multiple operators.
- **operator = op** stores the selected operator.
- **display.value += op** shows the operator on the display.

calculateResult() Function

- This function runs when the = **button** is clicked.
- **if condition** checks whether num1, operator, and num2 exist.
- **parseFloat()** converts string values into numbers.
- **if / else if** conditions perform the correct calculation.
- **result** stores the calculated value.
- **display.value = result** shows the answer on the screen.

Resetting for Next Calculation

- **num1 = result.toString()** allows chaining calculations.
- **operator = ""** clears the operator.
- **num2 = ""** clears the second number.

clearDisplay() Function

- This function runs when the **C button** is clicked.
- **display.value = ""** clears the calculator screen.
- **num1, operator, num2** are reset to empty values.
- The calculator is ready for a new calculation.

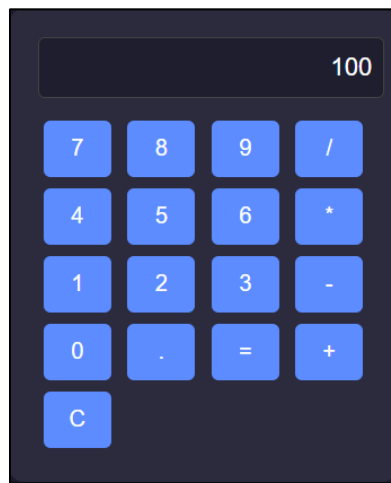
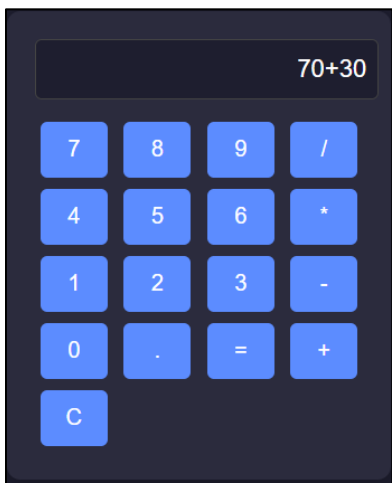
Final Conclusion

- This JavaScript code controls calculator operations.
- It handles **number input, operator selection, and calculation logic**.
- It avoids errors by checking conditions before calculation.
- The calculator performs basic arithmetic operations.

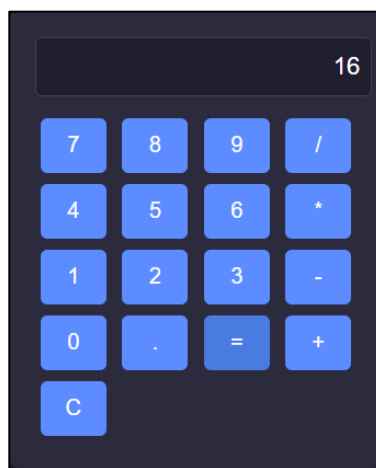
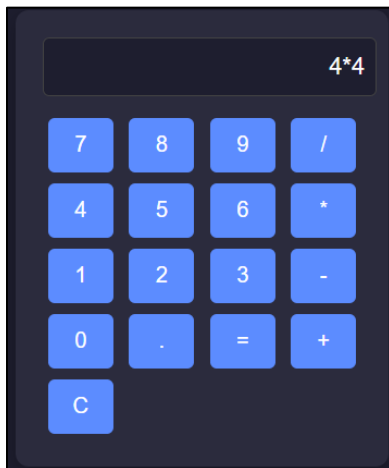
Output:



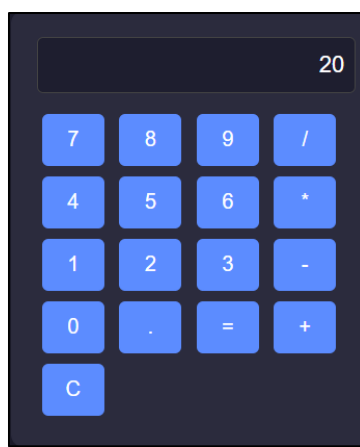
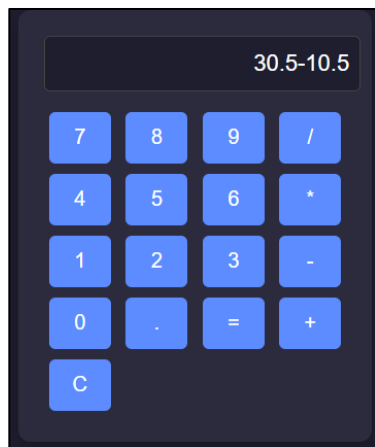
- Addition:



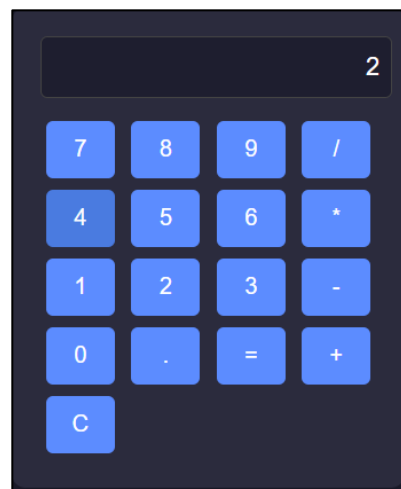
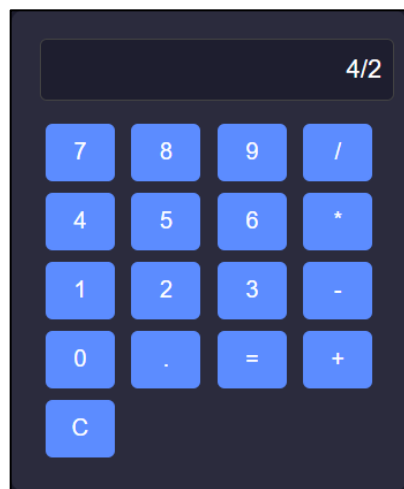
- Multiplication:



- Subtraction:



- Division:



✦ THE END ✦